

Analysis group meeting

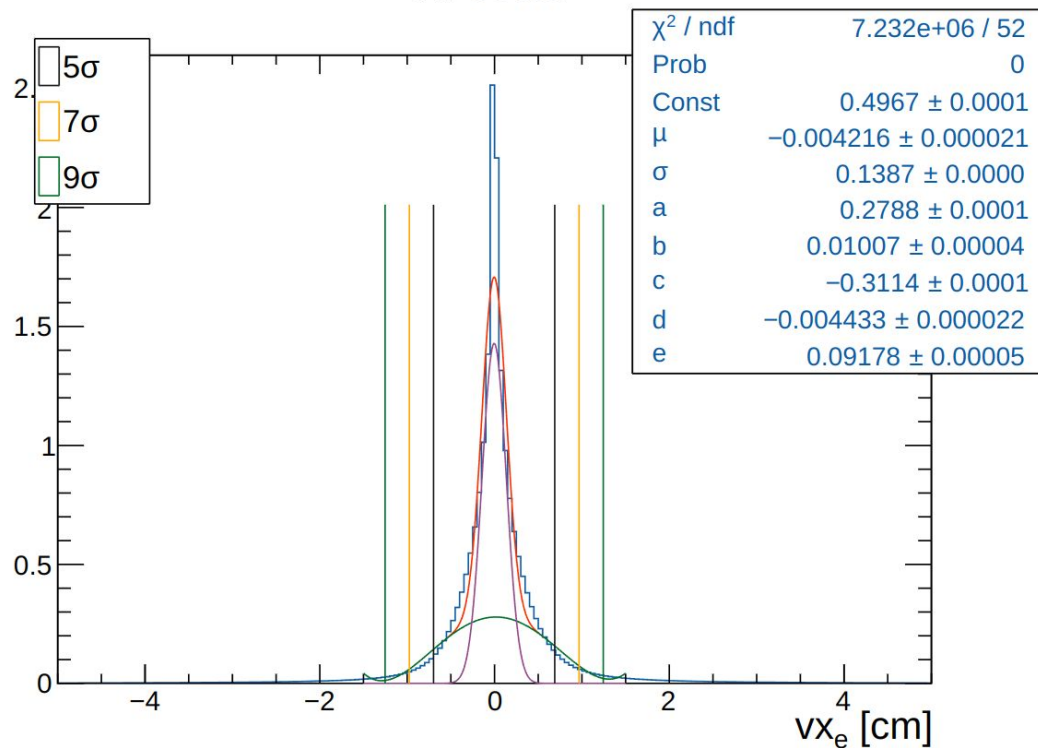
Mathieu Ouillon (Mississippi State University)

November 05, 2024



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UNIVERSITY™

All Sector

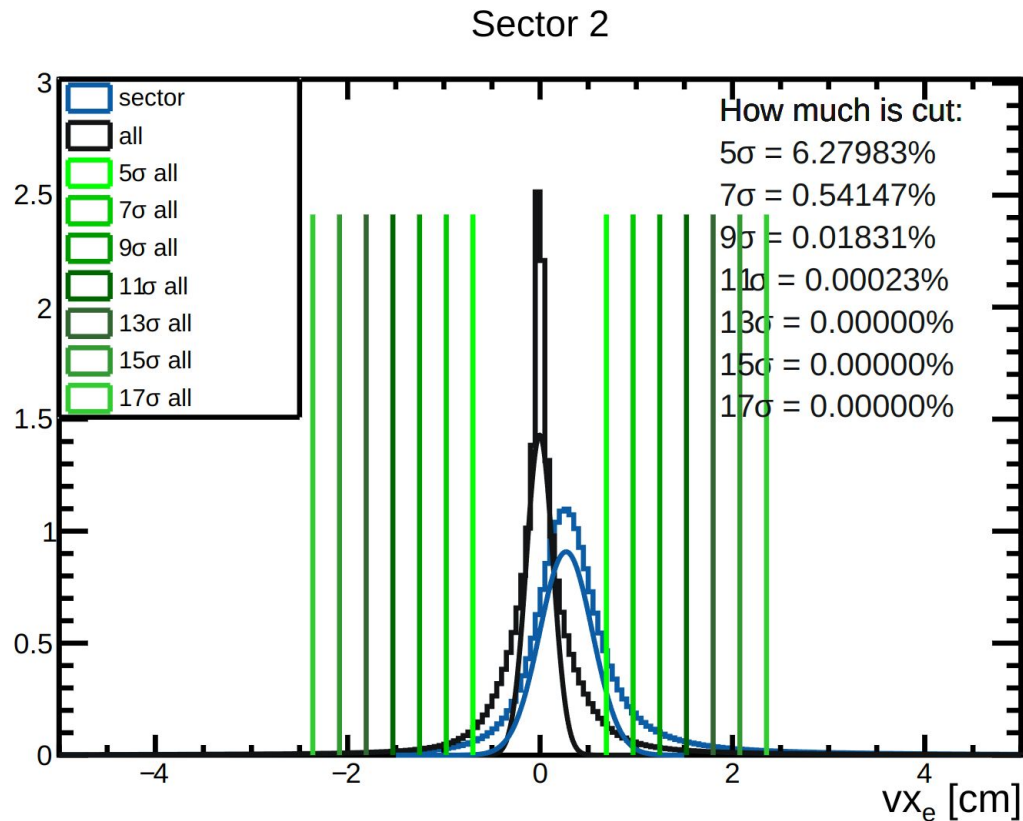


data: all pass0v6

target: CuSn

Fit with a gaussian + 4 order polynomial.

RG-D: Vertex study



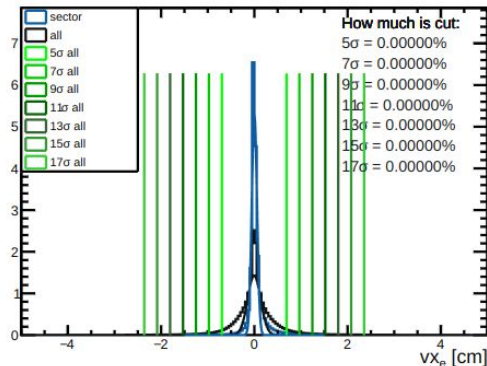
data: all pass0v6
target: CuSn

Fit sector 2 with a gaussian + 4 order polynomial.

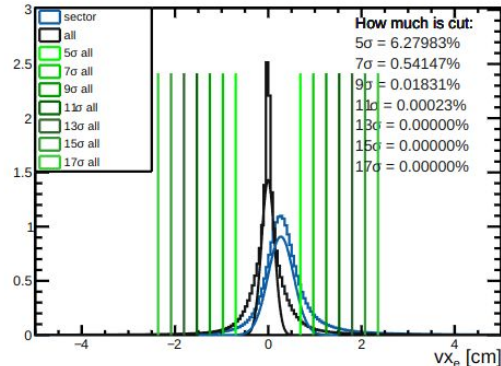
We look at how much of the sector's gaussian is cut by applying the cuts at 5, 7,... σ of the gaussian obtained with all sectors.

Sector 2 is the worse one. And a cut at 9σ keep most of the stats

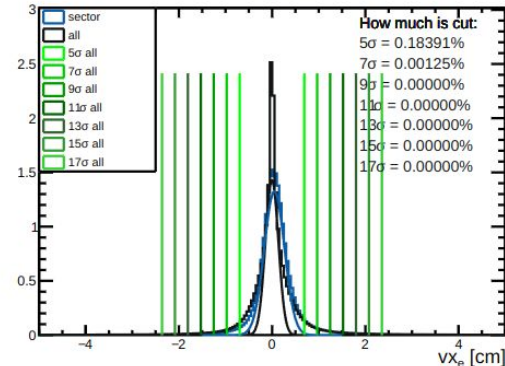
Sector 1



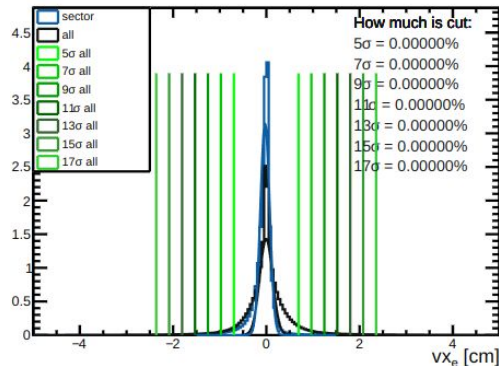
Sector 2



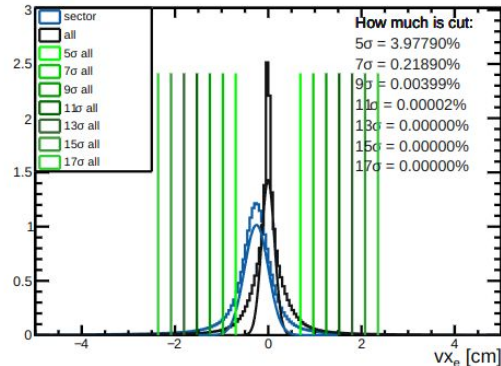
Sector 3



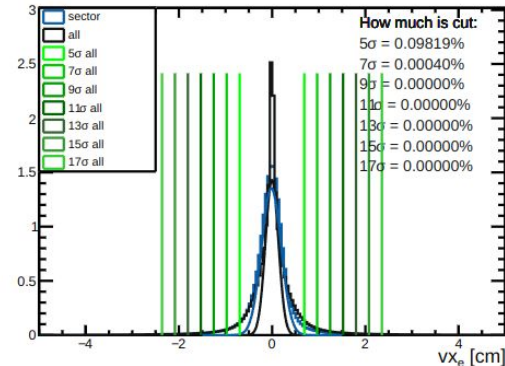
Sector 4



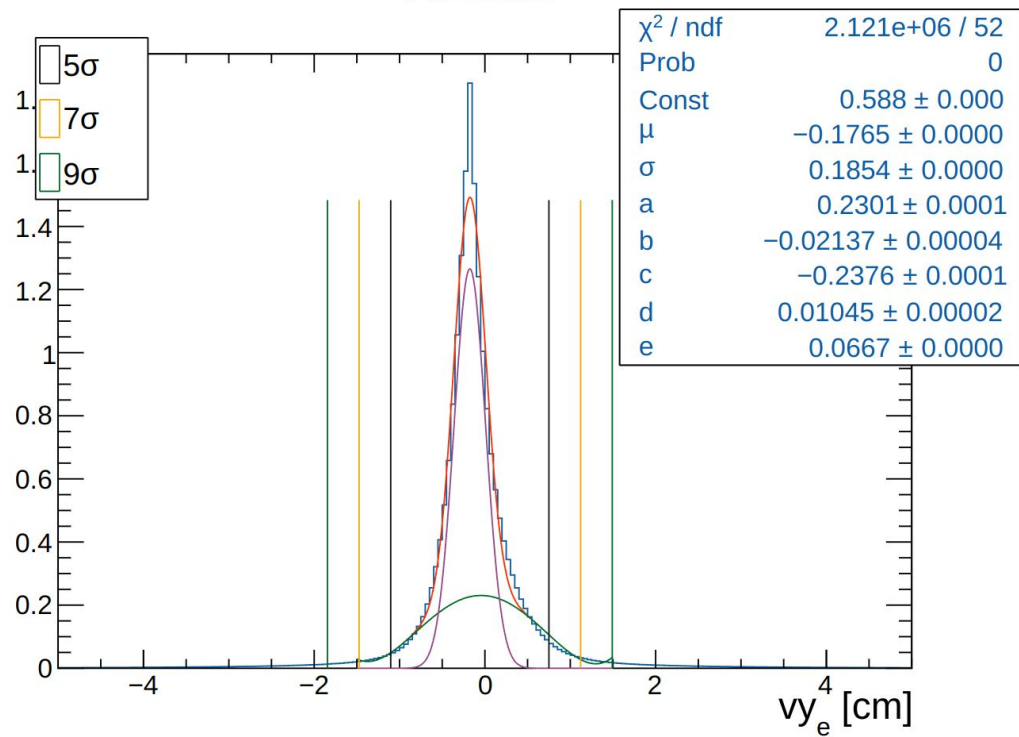
Sector 5



Sector 6



All Sector



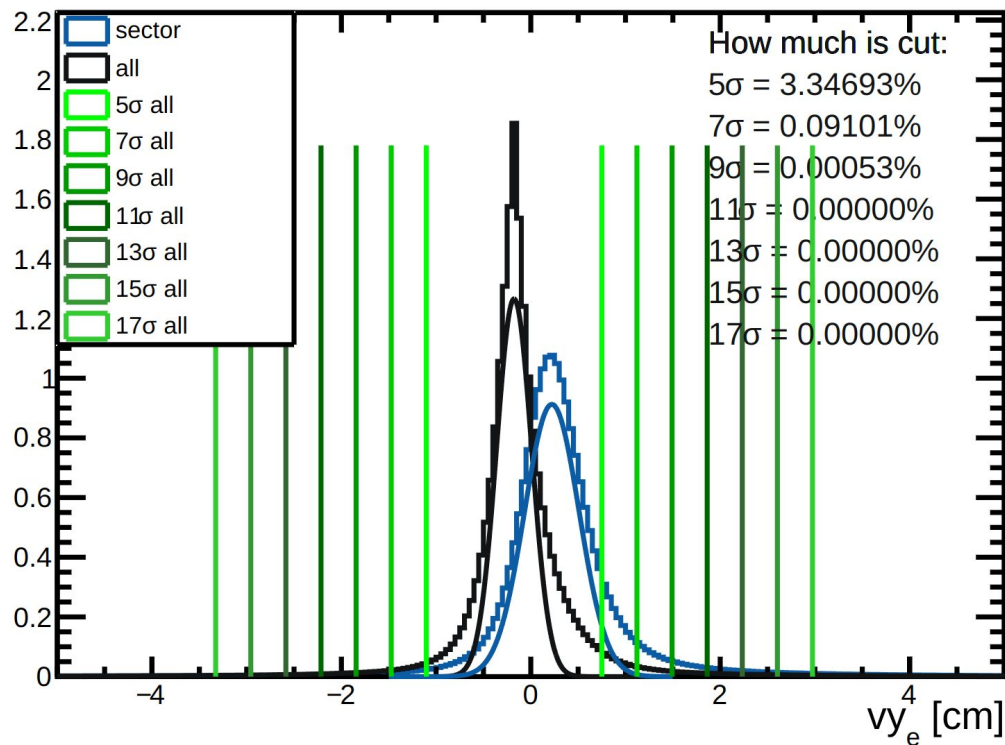
data: all pass0v6

target: CuSn

V_y is fit with a gaussian + 4 order polynomial.

RG-D: Vertex study

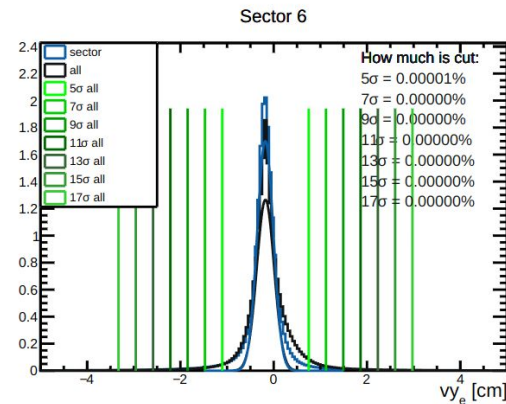
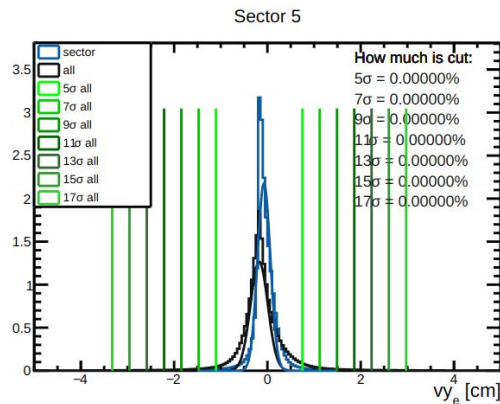
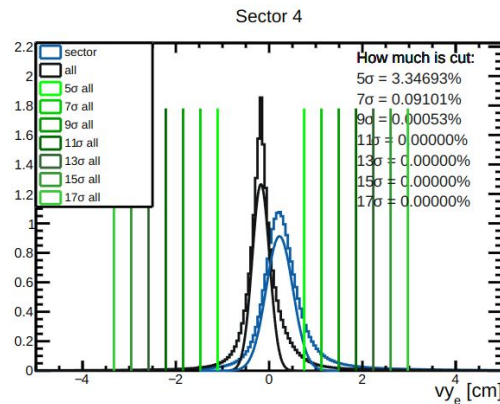
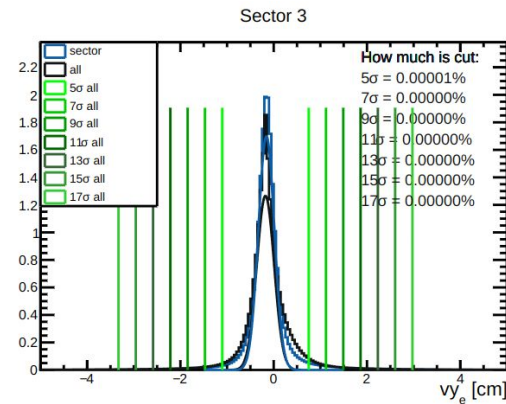
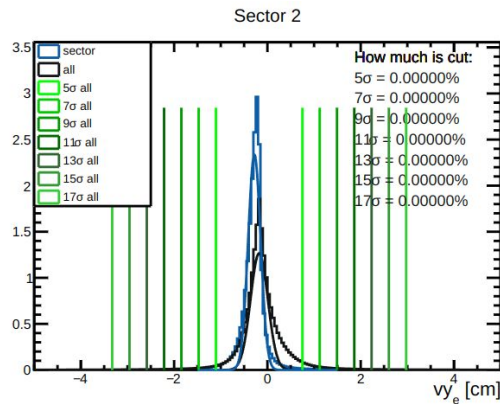
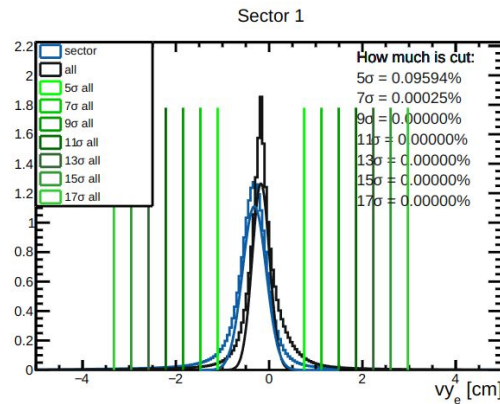
Sector 4

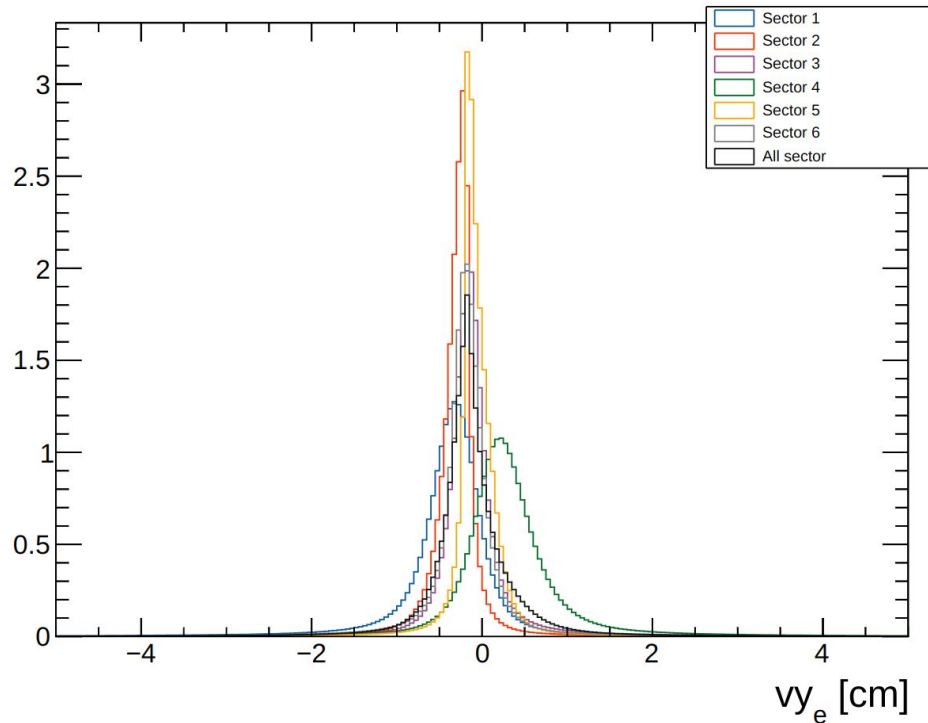
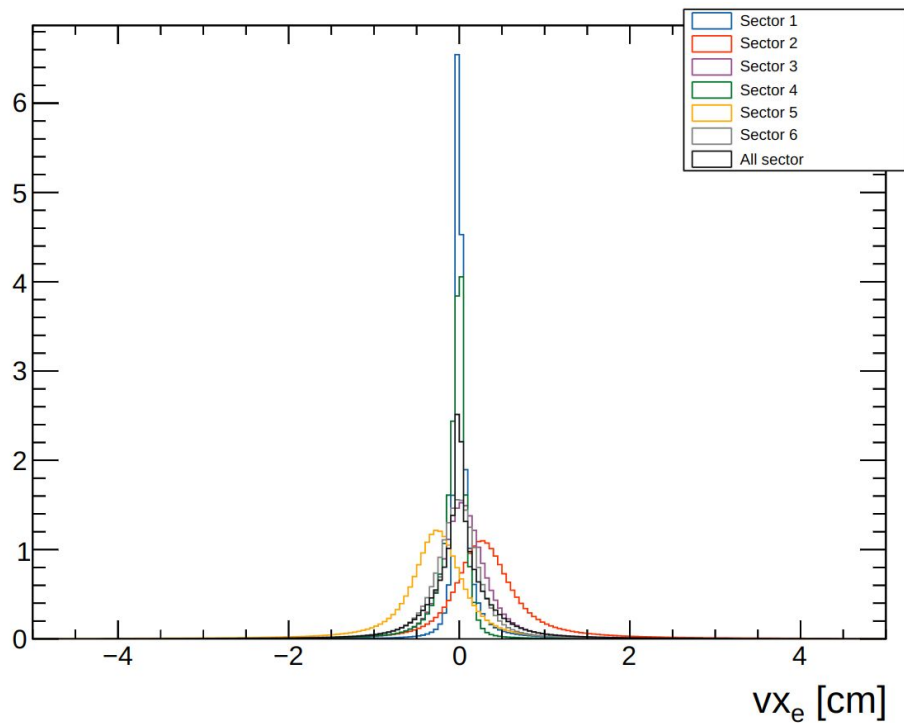


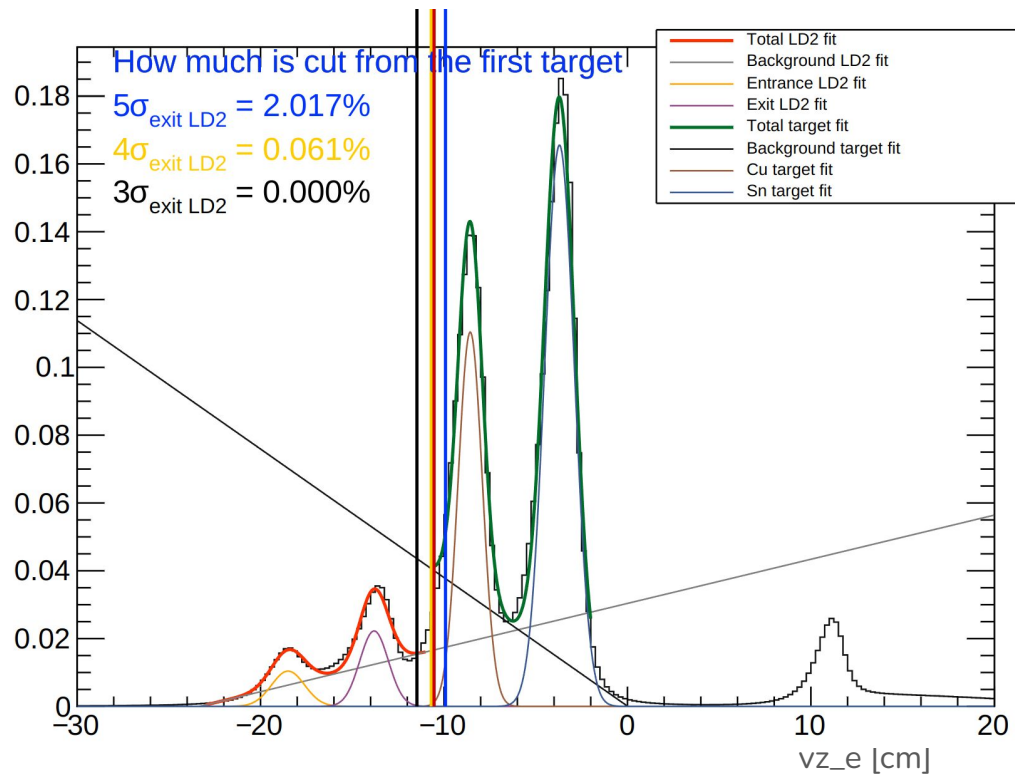
Fit sector 4 with a gaussian + 4 order polynomial.

We look at how much of the sector's gaussian is cut by applying the cuts at 5, 7,... σ of the gaussian obtained with all sectors.

Sector 4 is the worse one. And a cut at 7 σ keep most of the stats







v_z distribution after a cut at 9σ on v_x and 7σ on v_y .

Two separate fits:

- The LD2 region: 2 gaussians + pol1
- The CuSn region: 2 gaussians + pol1

Same study for carbon target

RG-D: Nuclear transparency

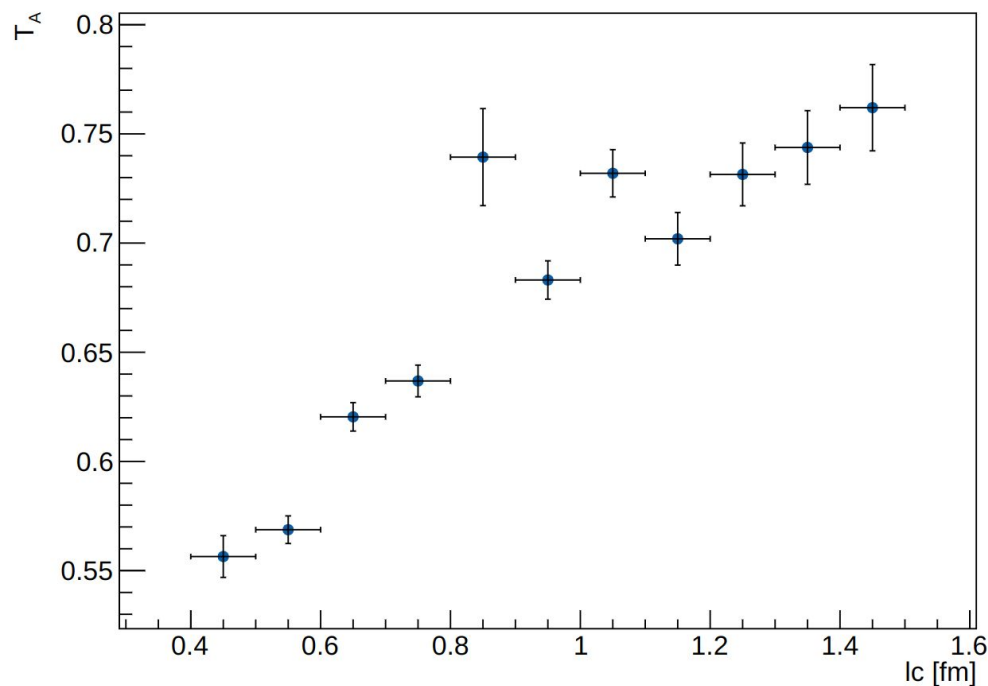
Nuclear transparency as function of l_c .

Data: Bspot carbon and LD2.

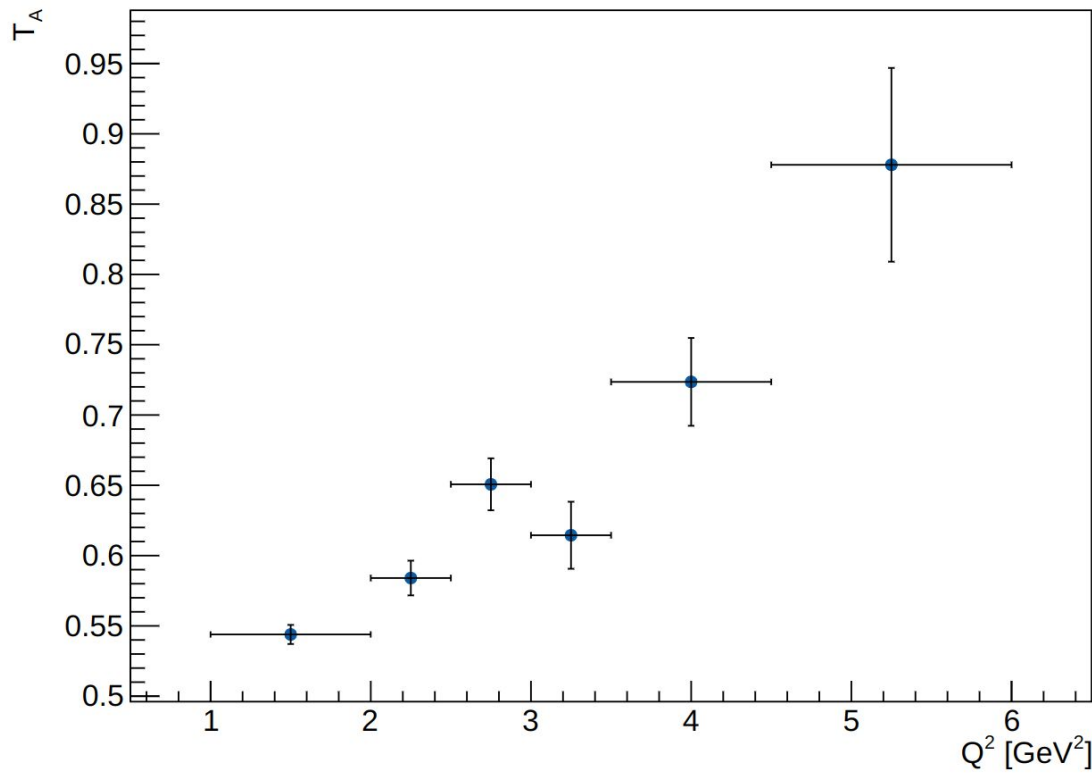
All inbending and outbending.

Process 354 M events per target.

- Cuts: electron:
 - $\text{pid} == 11$ and $\text{status} < 0$
 - $-12 < v_z < 5$ cm
 - $-5 < \text{chi2pid} < 5$
- pion minus/plus:
 - $-5 < \text{chi2pid} < 5$
- $W > 2$ GeV
- $z_h > 0.9$
- $0.1 < -t < 0.5$ GeV²



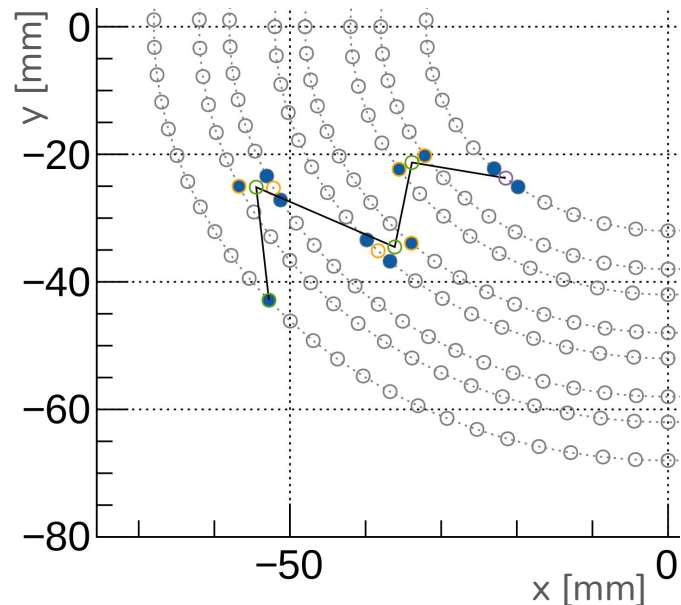
Same cuts + $l_c < 0.6$ fm



- Have discussed with Gagik:

- Compute the purity: number of good tracks classified as good / number of tracks classified as good vs. the threshold.
- Add other informations such as angle between two points.
- To implement the model in Java we can use the library [Deep Java Library \(DJL\)](#).
- We can also try different parameters of the position: r and ϕ , or just a wire number (something like $\text{superlayer} * 1000 + \text{layer} * 100 + \text{wire number}$).
- Otherwise, it looks good.

For the j4ml/j4np library that is used for AI for the DC, it is used under the hood the [Deep Netts](#) library which is not a free library (licensing issue).

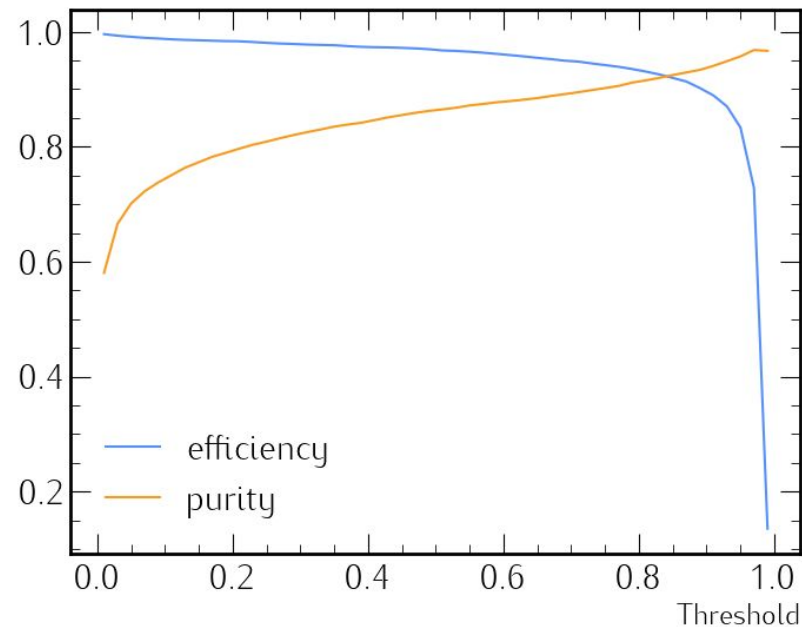


Change the definition of efficiency to better match what is use in others presentation and to have a efficiency of 100% at a threshold of 0.

efficiency = number of event where the model find at least the good track / number of event.

Plot purity and efficiency as function of threshold.

The threshold is a value above which the model's output is considered to be 1



Luminosity: $2 \cdot 10^{35} \text{cm}^{-2} \text{s}^{-1}$, for 55 PAC days

That give a integrated luminosity of $4.7 \cdot 10^{41} \text{cm}^{-2}$

TOPEG given the cross section in nb^{-1} : $1 \text{nb}^{-1} \mapsto 10^{33} \text{cm}^{-2}$

So the integrated luminosity is $4.7 \cdot 10^8 \text{nb}^{-1}$

Model 361 $\rightarrow$ Coherent He-4 DVCS neglecting the Real part of H

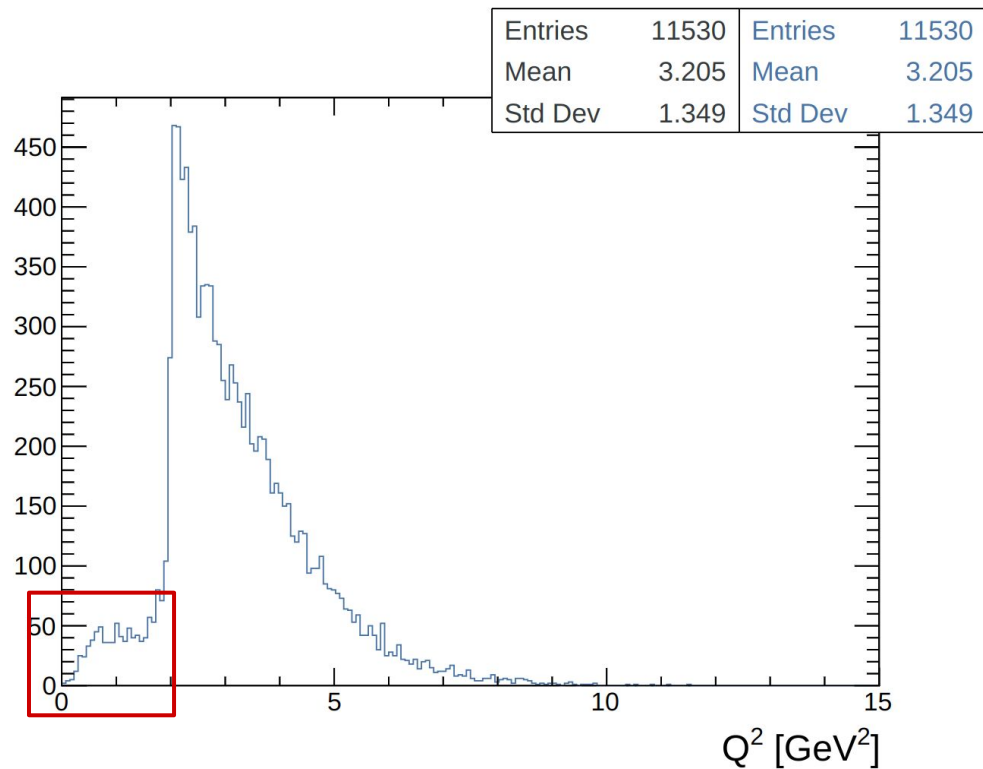
Cross section given by the event generator: 424.3 nb

$$N_{event} = L_{int} \cdot \sigma = 4.7 \cdot 10^8 \cdot 424.38971 = 1.9946 \cdot 10^{11}$$

Start from config file from myurov:
/work/clas12/myurov/temp/jlab22_c
ode/lund/root_config.txt

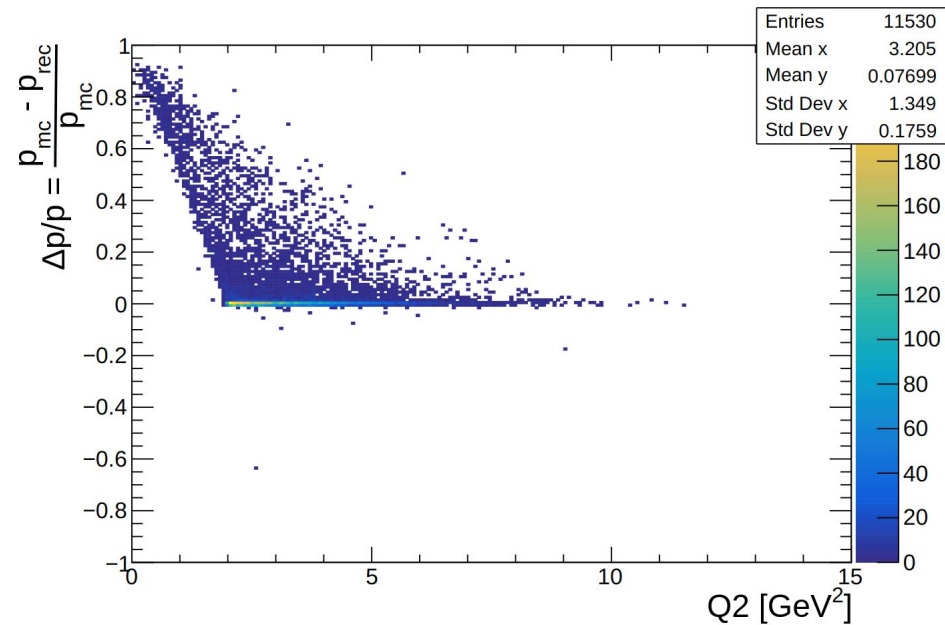
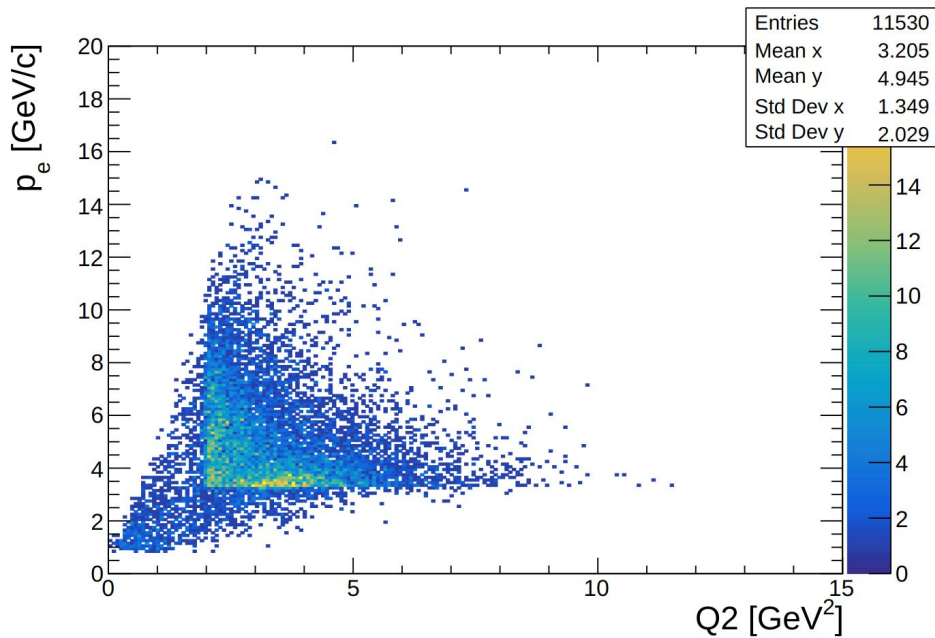
Simulate 366 000 events,
outbending configuration.

TOPEG Configuration Model 361	
10000	Nb of TFoam cells
5000	Nb of samples per cell
22.0	Electron beam energy (GeV)
0.0	Nuclear beam momentum (GeV)
0.15	y min
0.85	y max
2.0	Q2 min (GeV ²)
12.0	Q2 Max (GeV ²)
16.0	W2 min (GeV ²)
0.7854	theta Max electron (rad)
0.01	t min (GeV ²)
1	t Range (GeV ²)
1.5	xBj max
-1	Electron beam helicity



electron cuts:
vz_min = -25 cm
vz_max = 25 cm
chi2_min = -5
chi2_max = 5

photon cuts:
E = 3 GeV
beta_min = 0.9
beta_max = 1.1

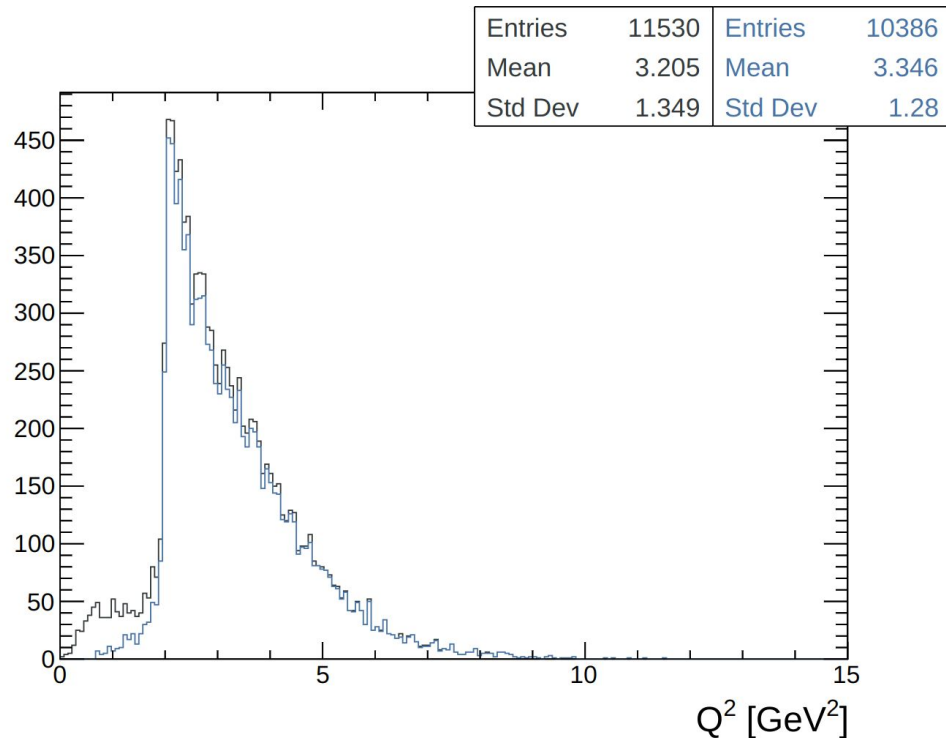


Cut: $p_e > 3 \text{ GeV}/c$

The cross section given by TOPEG for
free proton with $x_b \ 0.2 \ 0.27 \ q^2 \ 2.5 \ 3.2$
 $y \ 0.05 \ 0.9 \ w_2 \ 10 \ t \ 0.25 \ 0.4 \ th \ 0.01$
 $0.745 \text{ beam } 11.0 : 226\text{nb}$

The cross section given by DVCSgen
for free proton with similar
kinematics is 1.3pb

For RGA_DVCS_xsec_analysis_note
the cross section for this bin is 1.26pb



Cut: $p_e > 3 \text{ GeV}/c$

The cross section given by TOPEG for
free proton with x_B 0.2 0.27 Q^2 2.5 3.2
 y 0.05 0.9 w_2 10 t 0.25 0.4 th 0.01
0.745 beam 11.0 : 226nb

The cross section given by DVCSgen
for free proton with similar
kinematics is 1.3pb

For RGA_DVCS_xsec_analysis_note
the cross section for this bin is 1.26pb

